

drowned human bodies is one of the major missions of rescuers. Men and machines work together to rescue such people at the earliest. However, due to shortage of manpower, lack of training and professional equipment, body-recovery missions become difficult. Resources or techniques available to find people under heaps of debris or in a huge water body are generally not smart or capable enough to locate people in the shortest time possible.

The possibility of human fatalities has inspired many recent technological advancements in the expanding field of human safety. In an industry that has long relied on trained dogs and/or human trainers/lifeguards, products like electronic surveillance and sensing/detection in every imaginable form are making their debut on the market, all with the unifying goal of rescuing the victims at the earliest.

Electronic sensing devices are being advocated strongly to be more reliable, effective, cheap and user-friendly to locate people buried under rubble and vast water bodies. Such devices lead the rescue team quickly and efficiently to the victim, saving precious minutes of fruitless search. Innumerable technologies available today have enhanced the safety to the full range of senses. These include a surveillance system that never blinks and sensors that can hear a cry for help, which is never uttered to 3D imaging that quite literally transcends both time and space. Here are some upcoming electronic sensing techniques to locate buried and drowning people.

Trapped person locator (TPL). TPL is an extremely efficient and sensitive rescue tool that uses leading edge technology. It is used for technical rescue operations using seismic technology to listen for signs of life. The device detects the presence of buried victims under rubble and identifies their exact location using wireless and/or wired ultra-sensitive seismic sensors.

Further, search allows two-way communication using its waterproof communication probe up to two metres with the victims once they are found. The detector is equipped with both seismic and acoustic sensors, which discriminate between environmental and relevant sounds. Wired and/or



Trapped person locator (TPL)

wireless seismic sensors can detect sounds of scratching, knocking, cries and other nearly-inaudible sounds made by trapped victims as they struggle to be rescued.

The system has adjustable noise reduction filters to eliminate unwanted low/high noises from pneumatic drills, vehicles, wind, electric power cables and so on. The communication probe's microphone and speaker help communicate with buried survivors. The detector is lightweight, durable and portable, and comes with a compact control box. Its interface allows detection of victims' location faster. The system has been used in the successful location and rescue of victims all over the world, and heralds a new era in the field of emergency equipment and rescue preparedness.

Electronic sensing of buried people. There are various scientific methods that can help locate buried victims. One of the most important methods is to identify and locate variations in the surface such as depressions or small hills, which might indicate that a body is buried underneath. A new device, however, uses a probe—slightly thicker than a human hair—to probe the soil, detecting ninhydrin-reactive-nitrogen (NRN) in its vapour phase that collects in air pockets around grave soil.

The probe consists of an alumina-coated, porous layer, open tubular column, with a motorised pipette to pull in air samples from the soil at ambient temperatures. To probe the soil under a concrete slab (for NRN sampling), a small hole (about 3mm) is drilled in the slab. At present only

probing hardware is portable, but engineers are working on making the processing end portable too, so that the whole system can be used in the field.

Scientists have developed small and extremely sensitive gas sensors for acetone, ammonia and isoprene—all metabolic products that are emitted in low concentrations via human breath or skin. They have even combined NRN sensors in a device with two additional sensors for carbon-dioxide and moisture.

This is important as individual substances could come from sources other than human. However, a combination of sensors provides scientists with reliable indicators of the presence of people.

FINDER (finding individuals for disaster and emergency response). Microwave radar technology is sensitive enough to identify the unique characteristics of human breathing pattern and heartbeat from that of other living creatures. NASA has developed a microwave-based radar technology named FINDER to detect a human heartbeat buried beneath 9.14m (30-feet) of crushed material or hidden behind 6.09m (20-feet) of solid concrete, and from a distance of 30.5m (100-feet) in open space.

Several prototypes of FINDER have been tested and developed. Testing has proved successful in locating people buried in nine metres of mixed concrete, rebar and gravel rubble from a distance of over nine metres. The debris sometimes causes the radar signals to bounce back irregularly, making it difficult to interpret the signals accurately.

Another difficult task is to isolate the relatively weak signals of a heartbeat from noisy signals. However, this problem has been tackled to some extent by using advanced data processing systems to pick up faint signals. The advantage of this technology is to allow first responders to quickly detect if a living human is present in the debris and whether he/she is conscious, which can help rescuers decide the most efficient course of action.

Test results with the old version of FINDER have resulted in design